SOTERIA: Video-based Seizure Prediction

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***Abstract*—Generalized tonic-clonic seizures (GTCS) carry a serious risk of injury and sudden death, but the current video-based seizure literature is built almost entirely to retrospectively detect seizures already in progress rather than anticipate how they will evolve. We introduce SOTERIA, a strictly causal video-based system that estimates, during an ongoing seizure, progression toward a later annotated clonic onset: it anticipates clonic onset in 83.5% of onset-positive recordings, with a median 29-second warning among alarmed positives. SOTERIA combines frozen spatiotemporal appearance embeddings, a causal linear recurrent unit (LRU), and a forward-only hidden Markov model (HMM) alarm decoder. Every prediction at time t depends only on observations up to t, without access to future frames or the eventual outcome. We evaluate 210 recordings from 67 patients using patient-disjoint 10-fold cross-validation repeated across 15 paired seeds. SOTERIA obtains a rolling peak-risk ROC-AUC mean of 0.759—higher than those of the causal Transformer, TCN, LSTM, and GRU evaluated under the same stack. Backbone, objective, and decoder ablations then test which parts of the system affect discrimination and which alter the warning–false-alarm trade-off. These results suggest that non-contact video contains predictive structure for progression toward clonic onset and could extend this kind of forecasting to settings where electrophysiological (EEG) monitoring is impractical.**

***Index Terms*—Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, epilepsy, seizure prediction, video-based monitoring, causal sequence modeling, rolling prediction, hidden Markov models, linear recurrent unit.**

I. Introduction

Epilepsy is among the most common serious neurological disorders [1], [2], and for many patients the defining hazard is its unpredictability. Focal seizures (FOS) may begin with subtle behavioral change and later progress to tonic stiffening and rhythmic clonic convulsions [3], [4]. The same outward convulsive presentation can arise from generalized onset (generalized tonic-clonic seizures, GTCS), or secondarily, when a focal seizure spreads to engage both hemispheres, termed focal-to-bilateral tonic-clonic seizures [2]. An accurate warning before this progression creates the opportunity to summon a caregiver or begin a protective response before convulsive onset, which matters because GTCS carry a tangible risk of injury, aspiration, and sudden unexpected death in epilepsy (SUDEP), the leading cause of epilepsy-related mortality [5].

Seizure progression is clinically ordered but variable: early symptoms may be absent or unobserved, and motor signs can change as activity propagates [4], [6]. An alarm model may therefore reasonably encode forward progression without assuming every recording traverses the same visible stages.

The clinical value of a monitoring technology depends on which question it answers, a distinction reviews make explicit by separating seizure detection from seizure prediction [7]. Detection asks whether a segment of already recorded signal contains a seizure, a retrospective judgment made once all relevant evidence, including whatever makes the seizure unambiguous, already exists. Prediction asks whether a future seizure, or a seizure's future progression, will occur from the signal observed so far, so a genuine predictor must produce its estimate at time t from the observed history $\hat{y}_t = f(x_{0:t})$ alone—a generally more ambiguous prefix than detection has access to. A useful forecasting system must therefore act before the predicted clinical endpoint is observable.

Successful seizure forecasting has been demonstrated almost exclusively through EEG. A landmark first-in-human trial demonstrated prospective forecasting from intracranial recordings [8], while later work showed seizure risk follows circadian and multi-day rhythms detectable with wearables [9], [10]. Seizures are therefore at least partly anticipatable, but these modalities are invasive or impractical for continuous ambient monitoring, where unwitnessed seizures are most dangerous [5]. Contactless video is a scalable alternative: EEG requires specialized equipment, trained staff, and—in intracranial form—a surgical implant, tying its use to the epilepsy monitoring unit (EMU), whereas a conventional camera is inexpensive and deployable almost anywhere. Vision-based machine learning for epilepsy has advanced rapidly, but the literature has overwhelmingly emphasized detection [11].

II. Related Work

Video-based seizure analysis has progressed from qualitative motion description to deep appearance, skeleton, and attention models. Early work characterized seizure semiology from patient video [12]. GESTURES transferred spatiotemporal representations from human-action recognition and released one of the field's few public feature resources [13], [14]. VSViG adapted skeleton-based action recognition to epilepsy-monitoring-unit (EMU) video and introduced the corpus later released as WU-SAHZU-EMU-Video [15], [16]. Other works detect tonic-clonic seizures, model longer observed sequences, use cross-joint attention, or combine privacy-preserving visual signals with electrocardiography [17]–[20].

These works primarily address detection rather than prediction as defined in Sec. I, so their results are best interpreted within their respective detection protocols. The one exception is recent work forecasting seizures from a contactless camera

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via cross-species transfer learning [21], which explicitly targets prediction within a fixed future horizon: both its inference procedure and training objective ensure the estimate at t depends only on $x_{0:t}$, and its evaluation does not reduce to whole-clip accuracy.

Data scarcity remains a central obstacle, since sensitive recordings and confidentiality agreements keep most datasets private. Two public datasets have nonetheless emerged: GESTURES distributes pre-extracted spatiotemporal features derived from raw video [13], [14], while WU-SAHZU-EMU-Video releases 33 recordings from 14 patients with facial regions masked [15], [16]. These make data sharing feasible, though feature-only data restricts encoder development and face masking removes part of the visual signal.

Thus, video prediction remains substantially less studied than detection, while scarce public data makes leakage-resistant evaluation especially important. Unlike EEG, video captures behavioral rather than neural activity and may ultimately serve as a complementary modality.

III. Methodology

SOTERIA is a model that estimates progression toward annotated clonic onset from video features observed during an ongoing seizure, with every prediction required to be computable at its stated decision time.

As discussed in Sec. II, we choose GESTURES over WU-SAHZU-EMU-Video for its larger release and its clonic-onset annotations; this choice inherits the visual-encoder and augmentation restrictions of frozen features noted there.

SOTERIA combines these frozen appearance embeddings with a causal Linear Recurrent Unit (LRU) and a forward-only HMM alarm decoder (Fig. 1). Each component has a distinct role: the LRU estimates current risk, while the HMM determines whether evidence is persistent enough to warrant an alarm. Separating these roles permits the backbone and alarm rule to be tested independently in ablations.

A. Causal Contract

We define an estimate to be causal if and only if every operation needed to produce it is available at its stated decision time. SOTERIA enforces this requirement across five boundaries:

1) Each GESTURES STCNN embedding is timestamped at its final source frame [13], so an embedding computed from frames $x_{t-7:t}$ is indexed at t and is not available before all source frames exist.
2) Onset-positive sequences are truncated before a fixed pre-onset exclusion boundary: a conservative margin against annotation-timing imprecision, not itself a computability constraint. This boundary is set using the clonic-onset annotation at data-construction time, not presented to the model at inference (Sec. III-C).
3) Each rolling window contains only embeddings whose timestamps do not exceed the current decision time.
4) Within each rolling window, both the LRU and HMM recurse forward without using a later token.
5) The evaluation clock advances independently of the eventual endpoint, so the prediction time is not chosen retrospectively from the outcome—a design choice that differs from [21].

Together, these conditions establish past-only replay on recorded data: the evaluator recomputes the preceding window at each update rather than carrying hidden state across the recording. This governs information availability at inference; throughput and live-stream behavior remain separate deployment questions. The causal constraint does not apply to training supervision, where eventual outcome and time-to-onset are targets rather than model inputs.

B. Dataset and Problem Formulation

We use the public GESTURES feature release [13], [14] for our training and evaluation. The archive contains 220 metadata rows and feature directories for 217 recordings. Six feature-bearing recordings contain *Discard=Yes*, leaving 211 usable recordings. One onset-positive recording has annotated onset at 0 s and therefore contains no eligible pre-onset history. Excluding it before fold construction yields 210 recordings from 67 patients: 76 onset-positive and 134 onset-negative.

The positive endpoint is annotated clonic onset. Seven otherwise eligible recordings have such an annotation despite *GTCS=False*; we retain them because clonic onset, rather than GTCS, is the modeled endpoint. Thus, the output cannot be interpreted as GTCS probability.

Some recordings contain synchronized small (S) and large (L) views. Both views are independent training instances so the model can use the released visual evidence, but calibration and evaluation deterministically use S when available and otherwise L in order to avoid double counting of one clinical event as two test outcomes. Patient grouping keeps every recording and view from one patient in the same partition, preventing direct sharing of patient appearance, environment, or semiology between training and test data.

The binary task distinguishes recordings that later reach annotated clonic onset from onset-negative focal-seizure record-

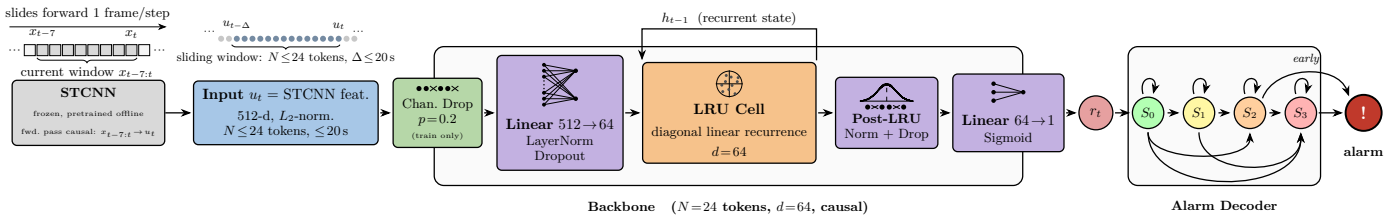


Fig. 1. Overall architecture of SOTERIA (Sec. III).

ings. At each nominal one-second update, the model receives at most the preceding 20 s, or all available history during warm-up, and emits one current risk. Positive replay stops before the exclusion boundary; onset-negative replay continues through the retained recording. The model is therefore evaluated repeatedly over each available history rather than on a retrospectively selected terminal window.

C. Feature Extraction and Preprocessing

GESTURES provides 512-dimensional spatiotemporal appearance embeddings from an STCNN pretrained on human-action recognition data [13], [14]. Because that pretraining precedes this task, it does not use this cohort's outcome labels. These frozen embeddings are SOTERIA's only input, and we L_2-normalize each token to reduce feature-scale variation while preserving its direction in the learned embedding space.

Each embedding summarizes eight frames. GESTURES indexes each snippet by its first frame [14], but we timestamp it by its last frame t, consistent with the causal notation of Sec. III-A: at 15 frames per second, the embedding spanning frames $t-7$ through t receives the causal timestamp $t/15$ seconds, once all eight source frames $x_{t-7:t}$ have been observed.

For onset-positive recordings, we retain only embeddings strictly earlier than 0.5 s before annotated clonic onset (Sec. III-A, boundary 2). This exclusion is a data-construction step: the clonic-onset annotation decides which snippets enter training, calibration, and evaluation, but the annotation itself is never presented to the model as a runtime input, consistent with the causal contract's treatment of outcome and time-to-onset as supervision targets rather than model inputs. This buffer separates every available snippet from the prediction endpoint. We exclude a recording if fewer than two embeddings remain, since causal sequence modeling requires both a history and a current token. Consequently, the final half-second and all post-onset video are absent from training, calibration, and evaluation.

During training, feature-channel dropout zeros each input dimension with probability $p = 0.2$ throughout the sampled window. Table IV measures its contribution.

D. Model Architecture

Every model input contains $N = 24$ tokens drawn from at most 20 s of history. A bounded recent-history window keeps the input and computational cost independent of total recording length while retaining enough context to represent changes developing over several seconds. The particular 20-s and 24-token settings are design choices and not independently validated clinical constants. Segment jitter varies sampled positions during training, reducing reliance on one deterministic alignment, whereas segment centers make evaluation reproducible.

During warm-up, the model receives all history available rather than waiting 20 s before it can warn. Histories containing fewer than 24 embeddings are resampled with repetition, and longer histories are divided into 24 temporal segments. The newest available embedding is forced into the final position so its output represents the current decision. This preserves early prediction opportunities but allows token spacing to reveal approximate elapsed history, which we quantify as a limitation. A learned projection compresses the 512-dimensional input to 64 dimensions; LayerNorm [22] and dropout stabilize and regularize the sequence representation. At token t, the deployed LRU accesses only tokens $[0, t]$, preserving causality.

1) Temporal Backbone: The temporal backbone is a single 64-dimensional LRU [23], part of the same state-space lineage as S4 and Mamba [24], [25], but implemented as a diagonal linear recurrence rather than a convolutional or selective-scan formulation:

$$\boldsymbol{h}_t = \boldsymbol{A} \odot \boldsymbol{h}_{t-1} + \boldsymbol{B}\boldsymbol{u}_t, \quad \boldsymbol{y}_t = 2\,\mathrm{Re}(\boldsymbol{C}\boldsymbol{h}_t) + \boldsymbol{D}\boldsymbol{u}_t \tag{1}$$

Its diagonal recurrence uses $A_k = \rho_k e^{\mathrm{j}\theta_k}$, where $\rho_k = \exp(-\exp(\nu_k)) \in (0, 1)$. The learned radii permit evidence retention over different timescales; we test the LRU empirically against causal GRU, LSTM, TCN, and Transformer backbones under otherwise matched conditions [26]–[29]. A linear head maps each LRU output to a risk estimate $p_t \in (0, 1)$.

2) Alarm Decoder: Token risk can fluctuate around an alarm threshold, so SOTERIA filters it with a four-state HMM [30]. This separates two questions: the neural model ranks current risk, while the decoder determines when accumulated evidence is strong enough to alarm. State S_0 is anchored to onset-negative risk, while S_1 through S_3 represent increasingly high-risk regimes and S_3 is absorbing. Four states provide a baseline regime and three levels of increasing risk. This is a modeling choice intended to give the filter more resolution than a two-state switch; the states are not assigned to, or validated as, clinical seizure phases.

The default transition matrix allows a state to remain fixed or move to any later state. Self-transitions let evidence accumulate within a regime, while the forward-only restriction encodes the general clinical expectation that progression toward the endpoint is directional. Skip transitions accommodate abrupt visible changes or an unobserved intermediate regime too brief to be resolved at the model's temporal resolution. These are structural priors rather than conclusions learned from this cohort, so the decoder ablation compares them with sequential, regressive, two-state, conservative firing, and raw-threshold alternatives.

Each r_t is modeled with a state-conditional Gaussian $\mathcal{N}(\mu_k, \sigma_k^2)$. The filtered posterior $\alpha_t(k) = P(S_t = k \mid r_{1:t})$ then follows the causal recursion

$$\alpha_t(k) \propto \mathcal{N}(r_t; \mu_k, \sigma_k^2) \sum_{j \le k} A_{jk}\, \alpha_{t-1}(j) \tag{2}$$

normalized so that $\sum_k \alpha_t(k) = 1$, where $A_{jk} = P(S_t = k \mid S_{t-1} = j)$ and the restriction $j \le k$ follows from the forward-only transition structure. Only $r_{1:t}$ enters this recursion, so decoding is causal.

An alarm fires at the first t with $\sum_{k \in \mathcal{F}} \alpha_t(k) \ge \kappa$ where $\mathcal{F} = \{S_2, S_3\}$. A separate calibration split selects the stay probabilities and threshold κ by maximizing F1 over 36 combinations. Calibration is necessary because sensitivity, false crossings, and warning time depend on the operating

point; keeping it inside the training partition prevents held-out test subjects from choosing that trade-off. The early criterion includes both high-risk states to favor sensitivity and warning opportunity, while the conservative criterion uses only S_3 to require stronger evidence.

Each fold contains only approximately 15 positive calibration recordings, making fully unsupervised emission estimates unstable. We therefore blend the Baum–Welch E-step responsibilities with soft time-to-onset targets centered 15, 8.25, and 1.5 s before annotated clonic onset. The resulting semi-supervised expectation maximization (SSEM) uses 0.7 posterior responsibility and 0.3 temporal supervision, preserving data-driven fitting while regularizing the ordering of the positive regimes. Forward–backward estimation follows the standard HMM formulation [30], and multistage seizure decoding has prior precedent [31]. All fitting remains inside the calibration split.

3) Inference Cost: When benchmarked on a single-core Apple M2, SOTERIA's entire pipeline has an estimated latency of 281–386 ms per update: 14.5 ms for HEVC decoding, 33 ms for preprocessing, 233–338 ms for the released STCNN [14], and 0.60 ms for the 24-token LRU.

E. Training

The backbone receives a recording-level binary target at every retained token: 1 for recordings with a later annotated clonic onset and 0 otherwise. Auxiliary time-dependent terms then shape how positive risk should evolve without changing that primary label.

The training objective is defined as

$$\begin{aligned}\mathcal{L} = \ & \mathrm{GDRO}(\mathcal{L}_{\text{focal}}) + 0.5\mathcal{L}_{\text{pair}} + 0.5\mathcal{L}_{\text{absolute}} \\ & + 0.3\mathcal{L}_{\text{hazard}} + 0.1\mathcal{L}_{\text{CVaR}}.\end{aligned} \tag{3}$$

Because the constituent losses use different reductions and numerical scales, their coefficients should not be interpreted as measures of relative importance.

- **Focal BCE:** Per-token binary cross-entropy is weighted by $(1 - p_t)^\gamma$ with $\gamma = 2$ [32], downweighting easy tokens and concentrating learning on ambiguous temporal evidence. Positive clips additionally receive a sigmoid time weight that rises from 0.30 toward 1.0, placing greater emphasis on observations closer to onset while retaining earlier context.
- **Group DRO:** Focal loss is first averaged within each view instance and then across instances for each patient. Patient weights are updated adversarially as $q_i \leftarrow q_i \exp(\eta \ell_i)$ with $\eta = 0.01$ [33], increasing the influence of patients with higher current training loss and reducing domination by easier patients.
- **Absolute margin ($\mathcal{L}_{\text{absolute}}$):** Pushes positive peak logits above 2 and negative peak logits below 2, encouraging a score scale compatible with a shared alarm threshold.
- **Pairwise margin ($\mathcal{L}_{\text{pair}}$):** A pairwise squared-hinge loss separates positive and negative peak logits by a margin of 1, aligning training with the recording-level peak score used for rolling discrimination.
- **Hazard:** To impose a time-dependent risk profile [34], positive clips are supervised with the rising target $z_t = \mathrm{sig}[-\rho(\tau_t - \tau)]$, where τ_t is time to onset, $\tau = 10$ s, and $\rho = 0.3$; negative clips use $z_t = 0$. It is a temporal shaping objective rather than a competing-risks estimator.
- **CVaR:** A conditional-value-at-risk term averages loss over the hardest 10% of onset-negative tokens [35], concentrating additional pressure on rare high-risk negatives most likely to produce false alarms.

Optimization uses AdamW [36] with learning rate 10^{-3} and weight decay 10^{-3}. Gradients are clipped at 1.0 to limit occasional large recurrent updates. Each fold is trained for 100 epochs and evaluated using its final weights, enforcing the same checkpoint-selection rule across seeds and ablations.

IV. Results

A. Evaluation Protocol

All results use patient-disjoint 10-fold cross-validation because recordings and views from one patient can share appearance, environment, and seizure semiology. Within each repetition, no patient appears in both training and testing. Twenty percent of the training subjects form a separate calibration split that fits HMM emissions and selects transition and alarm thresholds. Thus, test subjects do not set either model parameters or the operating point within that repetition.

We repeat the full procedure across the same 15 seeds for every condition, reused across conditions to create a paired comparison. Tables report the mean and 95% Student-t interval across seeds. Two-sided Wilcoxon signed-rank tests [37] compare paired seed-wise metrics, followed by Holm adjustment [38] within each ablation family and metric; this asks whether a change is consistent across randomized folds and initializations of this fixed cohort, not whether the 15 repetitions behave as independent patient samples.

For recording-level inference, we average each recording's peak risk across seeds and compare the correlated ROC curves with paired DeLong tests [39], Holm-adjusted within the architecture, BCE-only architecture, adjacent-objective, and complete-objective families. Because patients can contribute multiple recordings, we also resample the 67 patients with replacement for 10,000 replicates, retain all recordings belonging to each sampled patient, and recompute the AUC difference [40]. This patient-clustered interval asks whether a comparison is robust to patient composition, not whether it reflects demographic fairness or which patients benefit.

A component-level claim is strongest when its descriptive change and all three analyses agree. Decoder variants reuse neural risk streams, so DeLong does not apply there; we instead compare sensitivity, negative-recording crossing rate, and warning offset across paired seeds.

Replay begins after two embeddings are available. A one-second target grid selects the newest causally available embedding at each update, and the final eligible embedding is also evaluated when the grid does not land on it. ROC-AUC

measures ranking across all thresholds, while PR-AUC emphasizes the less frequent onset-positive class. Operational metrics use the first crossing per recording, which fixes the available warning time: positive-recording sensitivity, the fraction of onset-negative recordings that ever cross, and the median first-alarm offset among alarmed positive recordings (negative offset denotes warning before annotated clonic onset). Exposure-normalized false alarms cover only onset-negative seizure footage, not ordinary daily activity.

B. Main Results

TABLE I
PRIMARY ROLLING PREDICTION PERFORMANCE (10-FOLD PATIENT-DISJOINT CV, 15 SEEDS). ± IS THE 95% CONFIDENCE-INTERVAL HALF-WIDTH OVER SEEDS.

Metric	SOTERIA
ROC-AUC	0.759 ± 0.010
PR-AUC	0.605 ± 0.016
Balanced accuracy	70.7 ± 0.8%
F1	0.648 ± 0.007
Positive-recording sensitivity	83.5 ± 1.5%
Negative-recording false-alarm rate	42.1 ± 2.3%
Median first-alarm offset	−29.2 ± 1.2 s

Table I reports the default LRU with the early, skip-permissive SSEM decoder. Across 15 repetitions, rolling peak-risk ROC-AUC is 0.759, with every seed-specific AUC above 0.720. At the calibrated operating point, the decoder reaches 83.5% sensitivity and crosses in 42.1% of onset-negative seizure recordings, with a median first alarm 29.2 s before annotated clonic onset among alarmed positive recordings (missed recordings receive no artificial zero-second offset).

C. Architecture Ablation

TABLE II
BACKBONE ABLATION UNDER ROLLING REPLAY; 15-SEED MEANS ± 95% CI HALF-WIDTH.

Backbone	Peak AUC	F1	Pos. sens. (%)	Neg. FAR (%)
LRU (full)	**0.759±.010**	**0.648±.007**	**83.5±1.5**	42.1±2.3
Transformer	0.749±.009	0.623±.014	77.9±2.5	**40.9±2.3**
TCN	0.745±.008	0.624±.007	82.0±1.6	45.8±2.2
LSTM	0.731±.010	0.605±.011	81.7±2.0	50.0±2.5
GRU	0.718±.012	0.607±.008	81.1±2.2	49.0±2.1
LRU (BCE only)	**0.689±.012**	**0.582±.011**	**81.6±2.9**	56.0±3.1
LSTM (BCE only)	0.675±.009	0.569±.013	78.0±2.8	**54.4±2.4**
GRU (BCE only)	0.676±.011	0.576±.007	81.3±2.2	57.3±1.8

The upper block changes only the causal backbone while holding inputs, folds, seeds, objective, and decoder family fixed. The LRU has the highest mean AUC, exceeding the Transformer by 0.010, TCN by 0.014, LSTM by 0.028, and GRU by 0.041 (Table III). All four paired seed-wise differences remain significant after Holm correction, but the recording-level DeLong tests are not significant after adjustment and all patient-clustered intervals include zero. The LRU is thus the most consistent performer across this study's randomized pipelines, without established population-level superiority over each comparator.

The lower block restricts to the recurrent backbones (TCN and the causal Transformer are not recurrent and are omitted here), removes channel dropout, all auxiliary objective terms, and the HMM, then applies a calibrated threshold directly to

TABLE III
INFERENTIAL AUC COMPARISONS, LRU MINUS COMPARATOR (SEC. IV-A DEFINES ALL TERMS). FINAL COLUMN: PATIENT-CLUSTERED 95% CI FOR Δ_{rec}.

Comparator	Δ_{seed} $(p_{W,H})$	Δ_{rec} $(p_{D,H})$	Patient-clustered 95% CI
Transformer	0.010 (.0315)	0.004 (1.000)	[−0.033, 0.043]
TCN	0.014 (.0315)	0.004 (1.000)	[−0.039, 0.042]
LSTM	0.028 (2.44e−4)	0.023 (.993)	[−0.021, 0.063]
GRU	0.041 (3.66e−4)	0.021 (1.000)	[−0.023, 0.065]

BCE-trained models. The LRU again has the highest mean, by 0.013–0.014, so its descriptive ranking does not depend entirely on the specialized objective or decoder. However, these seed-wise differences do not survive Holm correction ($p = 0.073$ against LSTM, $p = 0.071$ against GRU), their DeLong-adjusted values equal 1.0, and their patient-clustered intervals include zero, so this simpler comparison preserves the descriptive ranking without providing independent evidence of superiority.

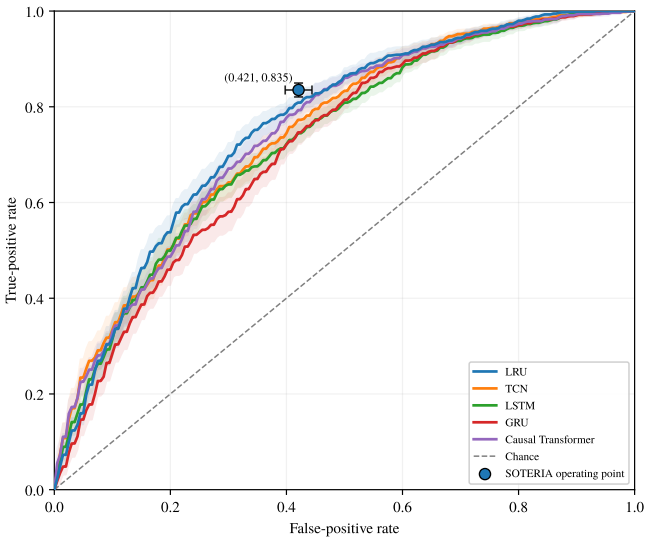


Fig. 2. Mean recording-level ROC curves under one-second causal rolling replay, averaged over 15 seeds; shading is the pointwise 95% Student-t interval across seeds. Marker: HMM decoder's operating point, not a point on any curve; error bars are the same 95% CI half-widths reported in Table I. Sensitivity at matched false-alarm rate is never below this curve (Sec. V-A).

D. Training-Objective Build-up

TABLE IV
CUMULATIVE CONTRIBUTION OF TRAINING OBJECTIVES UNDER ROLLING REPLAY; 15-SEED MEANS ± 95% CI HALF-WIDTH.

Objective build-up	Peak AUC	F1	Pos. sens. (%)	Neg. FAR (%)
Bare LRU (BCE + threshold)	0.689±.012	0.582±.011	81.6±2.9	56.0±3.1
+ Channel dropout ($p = 0.2$)	0.728±.008	0.593±.014	78.2±3.0	48.7±3.1
+ Hazard loss ($\lambda_2 = 0.3$)	0.739±.011	0.607±.011	75.7±1.8	41.9±2.9
+ Absolute margin	0.752±.010	0.614±.014	74.6±3.1	38.6±3.2
+ Focal weighting ($\gamma = 2$)	0.754±.009	0.617±.013	75.0±2.9	38.7±3.2
+ Group DRO ($\eta = 0.01$)	0.758±.007	0.617±.012	74.5±2.3	**37.9±3.0**
+ Pairwise margin ($\lambda_1 = 0.5$)	**0.762±.010**	0.613±.014	74.1±3.6	38.4±3.6
+ CVaR ($\lambda = 0.1$)	0.759±.010	0.616±.012	75.6±3.2	39.5±2.9
+ HMM decoder (full)	0.759±.010	**0.648±.007**	**83.5±1.5**	42.1±2.3

Table IV adds each component cumulatively, so an adjacent contrast measures one addition conditional on everything below it and on the chosen order. The complete learned-risk objective (CVaR row) raises mean seed-wise AUC from 0.689

to 0.759 relative to bare BCE, a 0.070 gain present in all 15 paired runs ($p_W = 6.10 \times 10^{-5}$). On seed-averaged recording scores the difference is 0.062 ($p_D = 0.0017$), with patient-clustered interval $[0.025, 0.101]$—three mutually consistent analyses supporting the combined objective as a whole.

TABLE V
INFERENTIAL AUC COMPARISONS FOR THE CUMULATIVE OBJECTIVE BUILD-UP; EACH ROW VS. THE ROW ABOVE, EXCEPT THE FINAL ROW (FULL VS. BCE-ONLY). SYMBOLS FOLLOW TABLE III.

Addition	Δ_{seed} ($p_{W,H}$)	Δ_{rec} ($p_{D,H}$)	Patient-clustered 95% CI
Channel dropout	+0.040 (.0013)	+0.030 (.554)	[−0.007, 0.067]
Hazard loss	+0.011 (.110)	+0.012 (1.000)	[−0.020, 0.046]
Absolute margin	+0.013 (.0129)	+0.014 (.404)	[−0.001, 0.030]
Focal weighting	+0.002 (.733)	+0.004 (1.000)	[−0.004, 0.015]
Group DRO	+0.003 (.733)	+0.004 (1.000)	[−0.005, 0.013]
Pairwise margin	+0.005 (.733)	+0.002 (1.000)	[−0.010, 0.015]
CVaR	−0.003 (.542)	−0.004 (.902)	[−0.011, 0.002]
Full vs. BCE-only	**+0.070 (6.10e−5)**	**+0.062 (.0017)**	**[0.025, 0.101]**

The adjacent results distinguish empirical support from design rationale. Channel dropout produces the largest mean increment, and the absolute margin also changes seed-wise AUC after Holm correction, but both patient-clustered intervals narrowly include zero, leaving their individual population-level effects unresolved. Hazard, focal weighting, Group DRO, and the pairwise margin produce positive but non-significant adjacent mean changes; CVaR produces a small negative one. These terms therefore remain parts of the evaluated recipe for their stated optimization roles, not separately established AUC improvements—notably, Group DRO is intended to emphasize high-loss patients, but this ablation does not show that the worst-performing patients improve. The HMM is fitted after training, leaves raw-risk AUC unchanged, and is evaluated separately below as an alarm rule.

E. Alarm Decoder Ablation

TABLE VI
ALARM-DECODER ABLATION ON FIXED RISK STREAMS. PARENTHETICAL VALUES ARE HOLM-ADJUSTED PAIRED WILCOXON p-VALUES VS. THE DEFAULT (ROW 1); "REGR." PERMITS REGRESSIVE (BACKWARD) STATE TRANSITIONS ON TOP OF THE DEFAULT'S SKIP TRANSITIONS. OFFSET IS RELATIVE TO ANNOTATED CLONIC ONSET.

Alarm mode	Pos. sens. (%)	Neg. FAR (%)	Offset (s)
HMM4-early-skip-SSEM	83.5±1.5	42.1±2.3	−29.2±1.2
HMM4-early-SSEM	**84.6±1.7** (.087)	43.4±2.1 (.514)	−28.3±1.3 (.980)
HMM4-early	76.0±2.1 (.005)	**34.4±2.7** (.005)	−25.3±1.7 (.007)
HMM4-early-full-SSEM	81.2±1.8 (.047)	39.2±2.6 (.035)	−27.3±1.0 (.014)
HMM4-early-skip-regr.-SSEM	82.7±1.6 (.151)	41.4±2.1 (.435)	−29.0±1.1 (1.000)
HMM4, highest-risk state only (skip)	78.1±2.1 (.006)	37.1±2.1 (.007)	−26.4±0.9 (.002)
HMM4, highest-risk state only (no skip)	78.3±2.2 (.006)	35.1±2.4 (.005)	−24.2±1.1 (.005)
HMM2 (two states)	80.3±2.3 (.047)	42.5±3.3 (.776)	**−29.5±1.5** (1.000)
Raw threshold on r_t	75.6±3.2 (.005)	39.5±2.9 (.435)	−29.1±1.3 (1.000)

The decoder ablation holds the trained risk streams fixed, so every difference is post-training. Relative to the optimized raw threshold, the default HMM raises sensitivity by 7.9 points ($p_H = 0.005$), F1 by 0.032 ($p_H = 0.012$), and balanced accuracy by 2.6 points ($p_H = 0.027$), while its increase in negative-recording crossings and its offset shift are not significant ($p_H = 0.435$ and 1.000). The default HMM thus improves calibrated sensitivity and aggregate alarm metrics consistently across seeds, without an established change in warning time or false crossings.

The remaining rows explain how that operating point is obtained. Requiring only the highest-risk state, and separately removing SSEM, both produce a statistically supported conservative shift: sensitivity and negative crossings fall and alarms occur later, supporting the value of temporal regularization for the selected early-warning behavior. Allowing regression and permitting skip transitions each leave the three primary metrics unchanged relative to their matched comparators; both are retained for directional clinical rationale rather than a demonstrated numerical advantage. Because the HMM thresholds a filtered posterior, these rows are distinct alarm rules rather than points sampled from the raw-risk ROC curve.

V. DISCUSSION

A. Why the Architecture Works

Across both the full-stack and BCE-only comparisons (Sec. IV-A), the LRU had the highest mean discrimination among tested causal backbones (Fig. 2 shows this across thresholds rather than at one alarm point), though its seed-wise advantage was significant only in the full-stack comparison, and patient-level resampling did not establish population-level superiority over any comparator in either. Its marked operating point lies near but not on the curve, since it comes from the HMM alarming on accumulated state evidence rather than a peak-risk cutoff (discussed below). Its compact recurrence and learned decay radii offer a plausible inductive bias for integrating gradual visual changes over multiple timescales, but the experiments evaluate the model rather than establish that mechanism as responsible.

The objective build-up supports a stronger conclusion at the recipe level than at the component level: the complete objective improves on bare BCE across all analyses, while individual components mostly retain only their design rationale, without independent population-level evidence of an adjacent gain.

Because the HMM operates after the neural risk stream, it changes the alarm operating point rather than raw-risk AUC. Consistent with this, a seed-paired check against the peak-risk threshold curve is informative: interpolating each seed's own curve at that seed's own decoder false-alarm rate, decoder sensitivity is never below the interpolated curve value across all 15 seeds and averages 3.0 points higher (95% CI [1.1, 4.9]; paired $t = 3.41$, $p = 0.004$; Wilcoxon $p = 0.008$), suggesting the decoder recovers sensitivity the instantaneous risk stream's own discrimination does not, at matched false-alarm rate.

The filtered posterior can also decrease even when every permitted latent path moves forward, since the model enforces ordered state transitions, not a monotone alarm score.

B. Clinical Applicability and Electrophysiology Complement

For a practical alarm, discrimination is only one part of performance. At the default operating point (Table I), the decoder alarms on the large majority of onset-positive recordings, with a median lead time of 29.2 s before annotated clonic onset. This window is defined relative to convulsive clonic onset, not seizure onset itself: SOTERIA does not claim to warn before the seizure begins, only that roughly half a minute of notice precedes the convulsive, injury-associated phase of a seizure already in progress. That distinction is what gives the

number practical weight: SUDEP and seizure-related injury are strongly associated with the convulsive and immediate post-ictal period—airway compromise, falls, and unwitnessed prone positioning—rather than with the earlier, less overtly dangerous phase of an ongoing seizure [5], [41]. A 29.2 s median lead time is on the timescale over which a nearby caregiver has been observed to respond to a tonic-clonic seizure in progress: in one epilepsy-monitoring-unit study, caregivers already present responded in a mean of 11 s from electrographic generalization, versus 22 s for hospital staff [42]. On this basis, the interval could plausibly allow a bystander to reposition a patient onto their side, clear nearby hazards, or alert a caregiver in an adjacent room before convulsions begin—the kind of intervention window nocturnal and ambient monitoring is intended to create for unwitnessed seizures [5], [41]. These observations establish the interval only as a plausible intervention window, not evidence that 29.2 s is sufficient to prevent injury or SUDEP; prospective validation is required. More conservative rules reduce false crossings at the cost of sensitivity and lead time, and resolving that trade-off likewise requires prospective data and clinical input.

SOTERIA's active trainable path contains 49,729 parameters, versus 63.7 million in the released frozen STCNN [14]: the sequence model is small, but the complete pipeline remains dominated by the frozen extractor.

These results place video alongside electrophysiology rather than against it. EEG measures neural activity and supports prediction over longer horizons, while video captures downstream behavioral progression and may be easier to deploy in wards or homes, so combining the two may be more useful than treating either as a replacement. The component-wise Apple M2 estimate is below the one-second decision interval, but clinical or embedded hardware has not been benchmarked.

Three onset-positive recordings reach annotated clonic onset within the first 20 s, leaving only 5–9 s of usable context; we retain them, since excluding these difficult early-onset cases would improve performance only through retrospective exclusion.

C. Limitations

The strength of each conclusion is bounded by the available evidence. First, as Sec. III-A establishes, rolling replay is retrospective and resets the model for each window despite its outcome-independent clock and past-only inputs.

Second, short histories are more common in the onset-negative class: 27.0% of rolling decisions use less than 20 s of available history (23.1% of positive decisions vs. 30.4% of negative decisions), and 40 recordings never reach a full 20-s history. Sampling 24 tokens across a variable-duration history changes their temporal density, so the model may infer elapsed history from embedding changes; this is past information rather than future leakage, but its association with the label may be dataset-specific. Requiring a full 20-s history would remove all 40 short recordings and the three difficult early-onset cases, so we retain early evaluation and disclose the cue.

Third, the cohort contains 210 recordings from 67 patients at one EMU. Patient-disjoint folds prevent direct overlap within each repetition, but the same cohort informed model development, including the 20-s and 24-token representation, so the headline estimate is internally cross-validated rather than obtained from an untouched external test set. The patient-clustered analyses show that several model-to-model differences remain uncertain at the patient level, and the available metadata cannot support demographic or clinical subgroup analysis; patient-disjoint splitting does not itself establish fairness across patients.

Fourth, the target follows the public release's clonic-onset annotations rather than being independently verified; the release does not resolve the seven recordings whose clonic-onset and *GTCS* fields disagree. Tonic-onset timing is unavailable, so the exclusion boundary guarantees separation from clonic onset but not necessarily from tonic activity.

Finally, onset-negative examples are focal-seizure recordings rather than ordinary activity, so the study does not measure alarms caused by sleep movement, coughing, caregivers, or occlusion. Frozen released features also prevent testing encoder improvements: with raw video access, alternative backbones and richer augmentations could plausibly improve on the frozen STCNN representation used here. SOTERIA remains a research model, not a validated medical device.

VI. Conclusion

This study tests causal video-based prediction of annotated clonic onset rather than retrospective seizure detection. Under patient-disjoint causal rolling replay, frozen appearance features discriminate recordings with a later annotated clonic onset, and controlled ablations give the LRU the highest mean AUC among tested causal backbones. The HMM then exposes, rather than removes, the trade-off between sensitivity, warning time, and false crossings, with a seed-paired check showing its sensitivity consistently at or above the raw risk stream's own curve at matched false-alarm rate. Wider public datasets are needed to test these findings beyond one EMU and cohort.

VII. Future Work

Future work should validate continuous multi-site streaming on long ordinary-activity negatives, calibrate clinically acceptable alarm trade-offs, and test fixed-cadence or variable-length inputs. Raw video access would also permit alternative visual encoders and augmentation in place of the released STCNN, while larger and external datasets, including subgroup-level analysis, are needed beyond this single-EMU cohort.

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